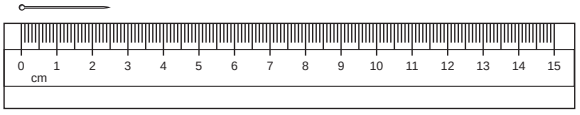
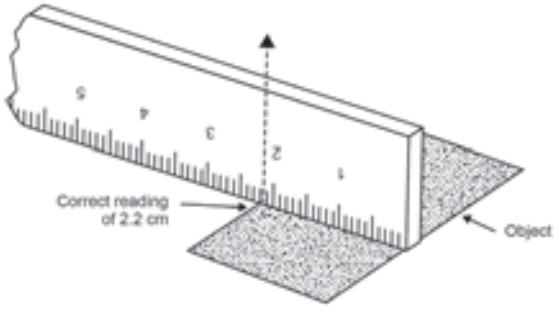
Corrective measures for Measurement

Measurement has to be accurate and the approach has to be correct always. In our day to day life approximation may not have much impact. But it has a large impact in scientific calculations. For example, if the curvature of key (lock and key) is changed even by 1 mm, the lock would not open. So, measurements have to be accurate in scientific calculations. Let us look at some common mistakes that occur while using a scale.

To measure the length of a pin



- The head of the pin has to coincide with '0' of the scale.
- Count the number of centimetre and from there count the number of finer divisions. The count of the division is in 'mm'
- In the above example the length of pin is 2 cm and 6 mm.
- Write the correct submultiple completely.



Note:

- Always keep the object parallel to the scale.
- Start the measurement from '0' of the scale.

Activity 2

Aim: To find the length of a curved line using a string.

Materials needed: A meter scale, a measuring tape, a string and a sketch pen

Method:

- Draw a curved line AB on a piece of paper
- Place a string along the curved line. Make sure that the string covers every bit of the curved line.
- Mark the points where the curved line begins and ends on the string.
- Now, stretch the string along the length of a meter scale and measure the distance between the two markings of the string and note it.
- This will give you the length of a curved line.



Find the length of a banana.



Parallax Error

Parallax is a displacement or difference in the apparent position of an object viewed along two different lines of sight.

Correct position of the eye is also important for taking measurement. Your eye must be vertically above the point where the measurement has to be